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1. Intelligent Agents in a Smart Home Automation System

A smart home automation system controls **lighting, temperature, and security** using sensor data such as motion, temperature, and time.

1. Simple Reflex Agent

- **Working:** Acts only on the current input using predefined rules.
- **Example:**
 - If motion is detected → Turn ON the light.
 - If no motion is detected → Turn OFF the light.
- **Advantages:**
 - Fast response.
 - Easy to implement.
- **Disadvantages:**
 - Cannot remember past events.
 - Cannot learn user preferences.

2. Model-Based Agent

- **Working:** Maintains an internal model (memory) of the environment and uses past information.
- **Example:**
 - Remembers that a room is occupied even if the motion sensor temporarily stops detecting movement.
 - Adjusts temperature based on previous readings.
- **Advantages:**
 - More accurate than a reflex agent.
 - Handles partially observable environments.
- **Disadvantages:**
 - More complex.
 - Requires memory.

3. Goal-Based Agent

- **Working:** Chooses actions to achieve a specific goal.
- **Example Goals:**
 - Maintain room temperature at 24°C.
 - Keep the home secure.
 - Turn off unnecessary lights to save electricity.
- **Advantages:**
 - Makes better decisions.
 - Plans actions to achieve goals.
- **Disadvantages:**
 - Requires planning.
 - Slower than reflex agents.

4. Utility-Based Agent

- **Working:** Selects the action that gives the highest overall benefit (utility).
- **Example:**
 - Balances comfort and energy savings.
 - If electricity prices are high, slightly reduces air conditioning while maintaining acceptable comfort.
 - Chooses the best security level without causing unnecessary alarms.
- **Advantages:**
 - Optimizes multiple objectives.
 - Provides the best overall performance.
- **Disadvantages:**
 - Most complex.
 - Needs a utility function and more computation.

Comparison

Agent Type	How it Works	Example in Smart Home
Simple Reflex	Uses current sensor input only	Motion detected → Light ON
Model-Based	Uses memory and current input	Remembers occupied room

Agent Type	How it Works	Example in Smart Home
Goal-Based	Works toward a goal	Maintain desired temperature
Utility-Based	Chooses the best overall action	Balance comfort, energy, and security

Best Choice

A **hybrid approach** combining **Model-Based** and **Utility-Based** agents is the most effective.

Reason:

- **Model-Based Agent**
 - Remembers user habits and room occupancy.
 - Uses past information for better decisions.
- **Utility-Based Agent**
 - Balances **comfort**, **energy efficiency**, and **security**.
 - Learns user preferences and chooses the most beneficial action.

Example:

- Learns that the user usually arrives home at **6 PM**.
- Starts cooling the room before arrival.
- Turns on lights only in occupied rooms.
- Activates security mode automatically when everyone leaves.
- Reduces unnecessary power consumption while maintaining comfort.

Conclusion

A **hybrid Model-Based + Utility-Based Agent** is the best choice for a smart home automation system. The model-based component remembers past events and user preferences, while the utility-based component selects actions that maximize comfort, minimize energy consumption, and improve security. This combination provides intelligent, adaptive, and efficient home automation.

2.Robot in an Unknown Maze

A robot is placed in an **unknown maze** and must find the exit without knowing the maze layout. It explores different paths until it reaches the goal.

1. Uniform Cost Search (UCS)

Working

- UCS expands the path with the **lowest total cost** first.
- It uses a **priority queue** to store paths based on their cumulative cost.
- It guarantees the **least-cost path** if all costs are positive.

Example

- If one path costs **5** and another costs **8**, UCS explores the path with cost **5** first.

Advantages

- Finds the **optimal (lowest-cost)** path.
- Works well when movement costs are different.
- Complete (finds a solution if one exists).

Disadvantages

- Uses **large memory**.
- Can be slow for large mazes.
- Explores many unnecessary nodes.

Complexity

- **Time:** $(O(b^{1+\lfloor C^*/\epsilon \rfloor}))$
- **Space:** $(O(b^{1+\lfloor C^*/\epsilon \rfloor}))$

where:

- **b** = branching factor
- **C*** = optimal path cost
- **ε** = minimum step cost

2. Iterative Deepening Search (IDS)

Working

- IDS repeatedly performs **Depth-Limited Search** with increasing depth limits.
- It starts with depth **0**, then **1**, **2**, **3**, and so on until the exit is found.

Example

- Search depth 0 → Not found.
- Search depth 1 → Not found.
- Search depth 2 → Exit found.

Advantages

- Requires **very little memory**.

- Complete for finite search spaces.
- Suitable when the goal depth is unknown.

Disadvantages

- Repeats searches from the root.
- Not optimal when path costs are different.

Complexity

- **Time:** ($O(b^d)$)
- **Space:** ($O(bd)$)

where:

- **b** = branching factor
- **d** = depth of the goal

Comparison of UCS and IDS

Feature	Uniform Cost Search (UCS)	Iterative Deepening Search (IDS)
Strategy	Lowest path cost first	Increasing depth limit
Complete	Yes	Yes
Optimal	Yes (for positive costs)	Only when all step costs are equal
Time Complexity	High	Moderate
Space Complexity	High	Low
Best Use	Different path costs	Unknown goal depth

Intelligent Agents

1. Simple Reflex Agent

- Acts only on the current sensor input.
- Example: Moves whenever an open path is detected.
- **Limitation:** Cannot remember previous locations.

2. Model-Based Agent

- Maintains an internal map of explored paths.
- Avoids revisiting the same locations.
- Best suited for an unknown maze.

3. Goal-Based Agent

- Plans actions to reach the exit.
- Chooses paths that move toward the goal.

4. Utility-Based Agent

- Selects the action with the highest overall benefit.
- Balances travel cost, time, and safety.

Effect of Agent Design on Decision-Making

- **Simple Reflex Agent:** Fast but may get stuck in loops.
- **Model-Based Agent:** Uses memory, making navigation more efficient.
- **Goal-Based Agent:** Plans systematically toward the exit.
- **Utility-Based Agent:** Chooses the best path by considering multiple factors such as distance, energy, and safety.

Best Combination

Model-Based Agent + Uniform Cost Search (UCS)

Justification

- The **Model-Based Agent** builds and updates an internal representation of the maze, preventing repeated exploration.
- **Uniform Cost Search** guarantees the **lowest-cost path** when movement costs differ.
- This combination improves navigation, reduces unnecessary revisits, and finds an optimal route.

If all movements have the same cost and memory is limited, a **Model-Based Agent + Iterative Deepening Search (IDS)** is also an effective choice because IDS uses much less memory while still finding a solution.

Conclusion (Exam Answer)

A **Model-Based Agent combined with Uniform Cost Search (UCS)** is the most effective solution for a robot navigating an unknown maze with varying path costs. The model-based agent remembers explored areas and avoids repeated exploration, while UCS guarantees the optimal path. If memory is limited and all moves have equal cost, **Model-Based Agent with IDS** is a good alternative because IDS requires less memory and remains complete.